

Hydrolysis of Ynamines

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You will be familiar with enamines as a source of nucleophilic carbon in organic synthesis. They can be hydrolyzed to give the ketone and amine in an acid-catalyzed pathway. This exploration features alkynamines, which are also subject to acid-catalyzed hydrolysis. We will explore evidence for a general acid catalysis mechanism and also consider kinetic isotope effects and substituent effects in our analysis. This exploration is based on work by Verhelst and Drenth.¹

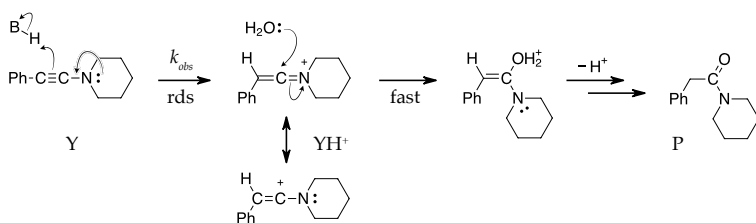
Ynamines

Ynamines are rapidly hydrolyzed in acidic water. The authors are investigating the rate of acid hydrolysis in basic solution ($pH > 7$) where there is very little hydronium ion to act as an acid. The only acids available will be water and the acidic form of a buffer. Is the acid catalysis specific or general? We can investigate. The authors use a series of N-substituted phenylacetylenes in this experiment. An example is shown in figure 1.

The phenyl ring provides a chromophore that is easily followed by UV/vis spectrophotometry. The concentrations of both the starting material and product can be easily measured. The reaction is rapid, even in basic conditions, and the rate was measured using stop-flow methods. Read the paper to experience some of the many challenges this caused for the authors.²

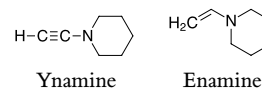
Reaction Mechanism

The mechanism is proposed to involve general acid catalysis of water addition. The acid could be hydronium ion, water or buffer.



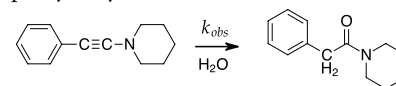
The rate-determining step is protonation to form the high-energy cationic intermediate. This cation is rapidly attacked by water to form an enol that tautomerizes to produce the amide product. The rate law will involve all the acids that can be involved in the protonation.³

$$\begin{aligned} v &= k_{\text{obs}}[\text{Y}] \\ k_{\text{obs}} &= k_{\text{H}_3\text{O}^+}[\text{H}^+] + k_{\text{H}_2\text{O}} + k_{\text{BH}}[\text{BH}] \end{aligned} \quad (1)$$



¹ "Addition of water to 1-alkynylamines. Kinetics and mechanism", W.F. Verhelst, W. Drenth, *J. Am. Chem. Soc.*, 1974, 96, 6692-6697. <https://doi.org/10.1021/ja00828a025>

Figure 1: Hydrolysis of 1-piperidino-2-phenylacetylene



² The reactant is poorly soluble in water and this required using an organic co-solvent. The mixing of water and solvent produced heat and there was no time to equilibrate the temperature in the flow cell. Read the paper to learn how this problem was solved.

Figure 2: General acid-catalyzed mechanism for ynamine hydrolysis

For a review of enamines as a source of nucleophilic carbon see [this chapter from a libretextbook](#). You may want to explore this nice list of definitions for [enols](#), [enolates](#) and [enamine nucleophiles](#)

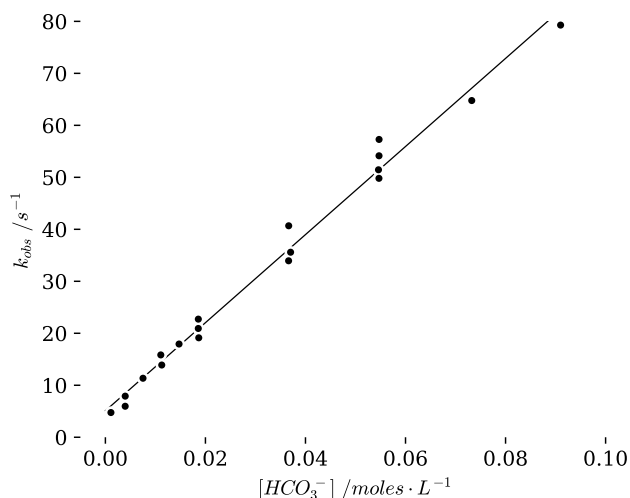
³ In the paper the authors treat the water rate as $k_{\text{H}_2\text{O}} = k_0[\text{H}_2\text{O}]$ and report the value of k_0 . I will use the more traditional treatment of $k_{\text{H}_2\text{O}}$ as including the concentration of water (55.5 mol L^{-1}) in this analysis of their work.

The hydronium rate is not significant in basic conditions. The observed rate constant is therefore...

$$k_{obs} = k_{H_2O} + k_{BH}[BH] \quad (2)$$

Catalysis

Buffer catalysis is observed in the reaction kinetics. The authors set the pH to the same value as the pK_a of the buffer. The buffer will be 50/50 in $[BH]/[B]$. The concentration of BH will be $\frac{1}{2}$ of the concentration of the buffer added. HCO_3^- has a pK_a value of 10.2. The authors measured the rate of hydrolysis of 1-piperidino-2-phenylacetylene with various concentrations of carbonate buffer at pH = 10.2.⁴



⁴ The data is found in figure 1 in the paper.¹ I used a graph digitizer application to pick the values of the points from the plot.

Figure 3: 1-piperidino-2-phenylacetylene hydrolysis with increasing concentrations of carbonate buffer. Data is presented in table 1

Table 1: Digitized data for hydrolysis of 1-piperidino-2-phenylacetylene in bicarbonate buffer from figure I of the paper.¹

$[HCO_3^-] / M$	k_{obs} / s^{-1}
0.0011	4.75
0.0040	5.96
0.0040	7.91
0.0075	11.3
0.0111	15.8
0.0113	13.9
0.0147	17.9
0.0186	20.9
0.0186	22.7
0.0187	19.1
0.0366	33.9
0.0366	40.7
0.0370	35.6
0.0546	51.5
0.0547	49.8
0.0547	54.1
0.0547	57.3
0.0732	64.8
0.0910	79.3

Fitting the data in figure 3 to equation 2, we find that $k_{H_2O} = 5.1 s^{-1}$ and $k_{BH} = 847 mol L^{-1} s^{-1}$. We can obtain the value for k_{BH} for different buffers this way and then make a Brønsted plot. The authors performed the hydrolysis using four buffers as the general base and also added a calculated k_{BH} value for water.

The water value was obtained by measuring the rate of hydrolysis in a high concentration of hydroxide where there is definitely zero contribution to the rate by the hydronium pathway. The k_{obs} for hydrolysis in 0.183 mol L⁻¹ NaOH is 1.03 s⁻¹. The base is water itself, so here we will divide by the concentration of water in water to get the k_{BH} for water as the general acid.⁵

$$k_{BH}^{H_2O} = \frac{1.03 s^{-1}}{55.5 mol L^{-1}} = 0.019 mol L^{-1} s^{-2}$$

⁵ The data for the water rate is recorded in table II of the paper.¹

Brønsted plot

Rate constants for the buffer general acid catalysts are reported for a series of buffer acids. The Brønsted equation relates the rate of buffer catalysis to the acidity of the general acid. Stronger acids can more effectively transfer a proton to the reactant. At a given concentration, the rate of general acid catalysis should increase with buffers of increasing acidity.

In the pre-equilibrium case of a specific acid, the increase in rate would be exactly the same as the increase in acidity. A 10-fold increase in acidity should give a 10-fold increase in the rate of buffer catalysis. A log-log plot would have a slope with a magnitude of one. If protonation is not important in the observed rate then the slope would be zero. But, if the proton is partially transferred in the r.d.s. (general acid catalysis), we would see a value between zero and one.

The Brønsted equation for acid catalysis is...

$$\log k_{BH} = \alpha \log K_a + c \quad (3)$$

Data for a series of buffer acids (including water as described above) is presented in table 2 and the Brønsted plot is shown in figure 4.

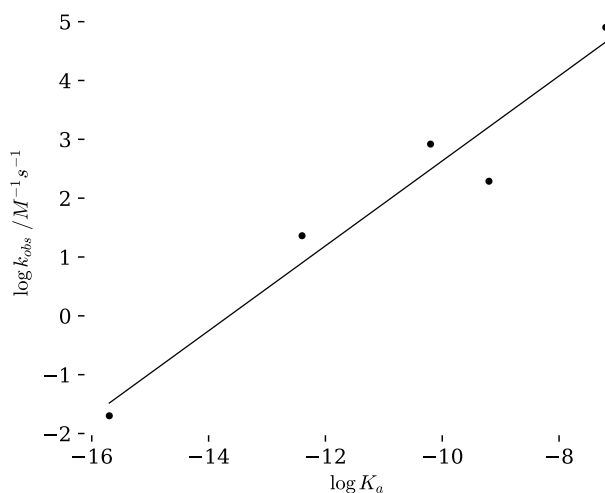


Figure 4: Brønsted plot for buffer catalyzed hydrolysis of 1-Piperidino-2-phenylacetylene

Table 2: k_{BH} values for different buffers.¹

Buffer	pK_a	$k_{BH} / M^{-1} s^{-1}$
$H_2PO_4^-$	7.2	80 000
HCO_3^-	10.2	830
$B(OH)_3(OH_2)$	9.2	194
HPO_4^{2-}	12.4	23
H_2O	15.7	0.019

The sensitivity factor, α , is 0.72. This indicates partial proton transfer in the transition state. The value is greater than 0.5 and this indicates a later T.S. as would be expected from a high energy intermediate according to the Hammond postulate. The authors obtained a value for α of 0.74. They statistically corrected⁶ the k_{BH} and K_a values according to the number of equivalent acidic proton and equivalent basic atoms in each base. For example, the acid $H_3PO_4^-$ has two acidic protons and the basic form has three basic oxygen atoms. These statistical corrections have small effects in most cases and you can see that my omission did not change the result much.⁷

⁶ "Statistical Factors in the Correlation of Rate Constants and Equilibrium Constants", S.W. Benson, *J. Am. Chem. Soc.*, 1958, 80, 5151-5154.
<http://doi.org/10.1021/ja01552a035>

⁷ In this case, applying these correction did not improve the line fit quality either.

Kinetic Isotope Effects

Kinetic isotope effects are also presented in the paper. The water rate was measured in basic water and basic deuterium hydroxide. The solvent kinetic isotope effect was found to be $k_{obs}^{\text{H}_2\text{O}}/k_{obs}^{\text{D}_2\text{O}} = 4.0 \pm 0.6$.⁸ This indicates that an O–H bond is being broken in the r.d.s. This is consistent with the idea of water acting as a general acid.

I would expect a similar primary kinetic isotope effect for the k_{BH} values of the four buffers as the proton would be quickly exchanged with deuterium in D_2O and the buffer–deuterium bond would be breaking during the r.d.s.

Taft Plot

In your reading in chapter 8 you would have encountered the Taft plot for correlating the electronic effect and the steric effect of substituents when they are close to the reaction centre (not spaced away by a benzene ring). The Taft plot is presented in equation 4. The electronic and steric substituent constants, σ^* and E_S , can be obtained from data tables. The corresponding reaction constants, ρ^* and δ , indicate the sensitivity of the system to the electronic and steric effects.

$$\log \frac{k_R}{k_H} = \rho^* \sigma^* + \delta E_S \quad (4)$$

The authors measured the rates of hydrolysis of a series of N-substituted 1-amino-2-phenylacetylenes. The steric term was considered to be similar for all the reactants used as they were all dialkylamines. No rate for the unsubstituted amine was reported. The authors combined the purportedly unchanging steric parameter with $\log k_H$ to give the following relationship.

$$\begin{aligned} \log k_R &= \rho^* \sigma^* + [\delta E_S + \log k_H] \\ \log k_R &= \rho^* \sigma^* + c \end{aligned}$$

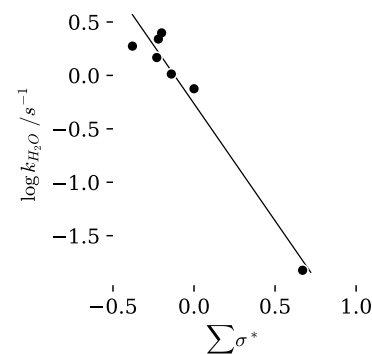
The data from the paper¹ is presented in table 3 and the plot is shown in figure 5. As you can see the plot, despite having seven samples, is a classic two-point-plot. A cloud of points is connected to a single far-flung point with a linear fit that has a slope of -2.0 . We can estimate that $\rho^* = -2.0$. Its not the best but it is clear that electron-withdrawing substituents reduce the rate of hydrolysis. This is consistent with the formation of a positively charged intermediate in the r.d.s.

⁸ see table II of the paper¹

Table 3: Reactants, substituent constants and k_{obs} values for the taft plot. From table III of the paper.¹

Reactant	$\Sigma \sigma^*$	$k_{\text{H}_2\text{O}} / \text{s}^{-1}$
<chem>Ph-C#C-N1CCOCC1</chem>	0.67	0.015
<chem>Ph-C#C-N(C)C</chem>	0	0.75
<chem>Ph-C#C-N1CCCCC1</chem>	-0.14	1.03
<chem>Ph-C#C-N(C)CC</chem>	-0.20	2.50
<chem>Ph-C#C-N(C)CC</chem>	-0.22	2.19
<chem>Ph-C#C-N(C)CC</chem>	-0.23	1.47
<chem>Ph-C#C-N(C)(C)C</chem>	-0.38	1.88

Figure 5: Taft plot



Conclusion

The Brønsted plot, Taft plot and solvent isotope effect are all consistent with the rate-determining-step being proton transfer from an acid to give a cationic intermediate.

Ynamines are useful reactants⁹ but were difficult to make and are very unstable. Recently, new synthetic methods have been introduced¹⁰ and stabilized versions have become commercially available as building blocks in organic synthesis.¹¹ There is much interest in these fascinating functional groups.

⁹ "Ynamine: A versatile tool in organic synthesis.", J. Ficini, *Tetrahedron*, **1976**, *32*, 1449-1486. [https://doi.org/10.1016/0040-4020\(76\)85200-3](https://doi.org/10.1016/0040-4020(76)85200-3); "Reactive species: Ynols and ynamines.", Y. Chiang, A.J. Kresge, S.W. Paine, V.V. Popik, *J. Phys. Org. Chem.*, **1996**, *9*, 361-370. [https://doi.org/10.1002/\(SICI\)1099-1395\(199606\)9:6<361::AID-POC793>3.0.CO;2-Y](https://doi.org/10.1002/(SICI)1099-1395(199606)9:6<361::AID-POC793>3.0.CO;2-Y)

¹⁰ "Copper-catalyzed domino synthesis of ynamines." M.L.N. Rao, S.S. Islam, P. Dasgupta, *Org. Biomol. Chem.*, **2021**, *19*, 7855-7860. <https://doi.org/10.1039/D1OB01383A>

¹¹ "A Journey in the Chemistry of Ynamides: From Synthesis to Applications." C. Zhang et al, *Chem. Lett.*, **2016**, *45*, 574-585. <https://doi.org/10.1246/cl.160260>